

Conditionals

Learning Objectives

- Understand if-then-else logic
- Use conditional blocks to make decisions in programs
- Combine conditions with sensing and operators

Engage (5-7 minutes)

A farmer checks the sky every morning. If it is cloudy, he carries an umbrella. If not, he does not. This is conditional thinking—making decisions based on conditions. Scratch conditionals let your sprites make smart choices too. Let us learn this decision-making skill.

Explore (10-12 minutes)

Drag an “if-then” block from Control. Put a sensing block inside like “touching mouse-pointer?” Add actions inside the if block. Try “if-then-else” to do one thing if true and another if false. Test it by making a sprite follow your mouse or hide when touched.

Explain (10-12 minutes)

Conditionals check whether something is true or false and respond accordingly. “If-then” runs actions only when the condition is met, while “if-then-else” offers two paths. You can combine multiple conditions using “and” and “or” operators. This makes your programs respond intelligently to different situations.

Elaborate (8-10 minutes)

Create a game where a sprite catches falling objects. Use “if touching edge” to bounce, “if key pressed” to move, and “if score equals 10” to show a win message. Like a traffic police directing vehicles based on signals, your sprites respond to different conditions.

Evaluate (5-8 minutes)

Exit ticket: Write a conditional statement using “if-then” for a real-life situation. What is the difference between “if-then” and “if-then-else”? Give an example of a condition you could check in Scratch.